

Control Valve Sizing

Valve position LAA01AA202 – circulation from the pressure tank back to the pressure tank

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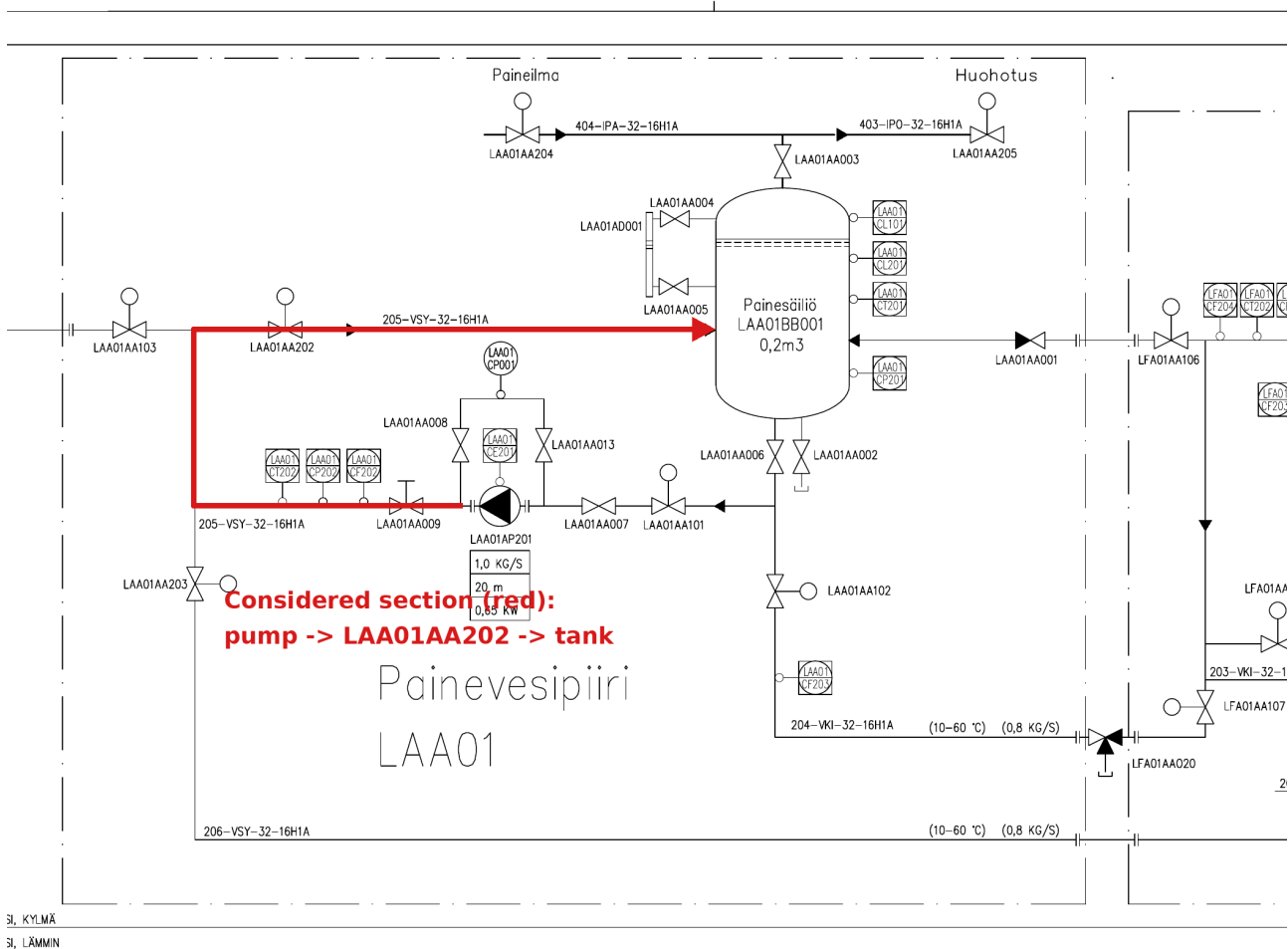
1 Introduction

The objective of this assignment is to carry out a complete control valve sizing exercise for a closed-loop water circulation process. The pump curve is evaluated, the pressure loss between the pump and the valve is calculated and from this the pressure drop across the valve and the process pressure ratio DP_m are determined. With these data a valve is sized in Valmet Nelprof. As required by the assignment, the sizing is first done with an RE-series segmented ball valve and then repeated with an M1-series ball valve, which also represents the type of the currently installed ball valve. The results are analysed on the basis of flow control theory (JAMK University of Applied Sciences, 2024; Valmet, 2022) and the more suitable valve is selected.

2 Process Application Description

The assigned valve position LAA01AA202 is part of a continuous closed-loop water circulation: water at 25°C is pumped from the 0.2 m³ pressure tank LAA01BB001 by pump LAA01AP201 and returned to the same tank through the control valve. Figure 1 shows the circuit in the P&I diagram. Following the flow, the water passes the manual valves LAA01AA006, LAA01AA101 and LAA01AA007 on the suction side, the pump, the flowmeter CF202, the pressure transmitter CP202, the temperature sensor CT202 and finally the control valve LAA01AA202 before it enters the tank. The section considered in the calculation runs from the pump discharge to the tank inlet and is shown schematically in Figure 2.

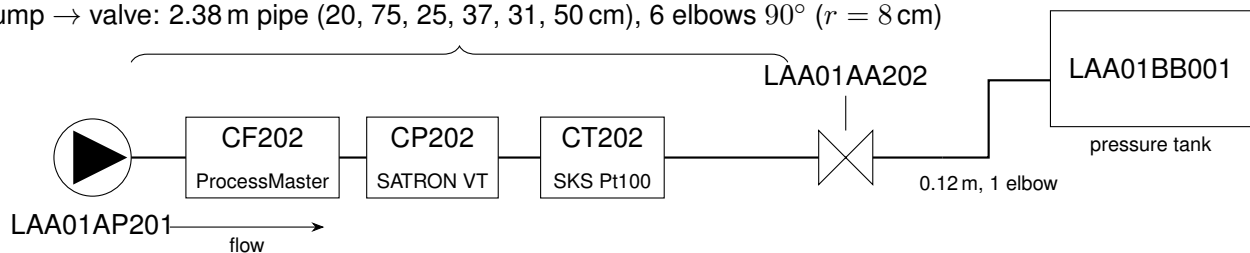
Figure 1. P&I diagram excerpt of the pressure water circuit LAA01 with the considered section marked in red



Note. Adapted from JAMK University of Applied Sciences (n.d.).

Figure 2. Schematic of the section pump → valve → tank (not to scale, instrument order as in the P&I diagram)

pump → valve: 2.38 m pipe (20, 75, 25, 37, 31, 50 cm), 6 elbows 90° ($r = 8$ cm)



Current installation (baseline):

- **Pump:** Kolmeks L-32A/2 centrifugal pump, 105 mm impeller, 3-phase motor, 50 Hz (3000 r/min), DN 32.
- **Current valve:** Jamesbury 25-716D-31-3600XTZ2 ball valve (DN 25, PN16), design pressure up to 500 kPa, design temperature up to 120°C, design flow range 0–0.8 kg/s according to the device description of LAA01AA202.

- **Current actuator:** AUMA electric part-turn actuator SGR 04.3.
- **Flow requirement:** design operating flow $q = 0.85 \text{ l/s}$ ($3.06 \text{ m}^3/\text{h}$, 3060 kg/h). This is about 6 % above the upper end of the documented design range of the current valve (0.8 kg/s).

3 Calculations and Sizing Methodology

3.1 Input Data and Assumptions

Table 1 lists the input data. The pipe size was not given in the assignment data and was determined from the P&I diagram.

Table 1. Input data

Quantity	Value	Source
Medium, temperature	Water, 25°C	Assignment data
Design flow q	$0.85 \text{ l/s} = 3.06 \text{ m}^3/\text{h}$	Assignment data
Density ρ / kinematic viscosity ν	1000 kg/m^3 / $0.893 \text{ mm}^2/\text{s}$	Water at 25°C (A3)
Vapour pressure p_v	0.0317 bar(a)	Nelprof (water, 25°C)
Pipe (pump to tank)	DN 32, $d = 32 \text{ mm}$, steel, $\varepsilon = 0.046 \text{ mm}$	P&I diagram (JAMK, n.d.)
Pump	Kolmeks L-32A/2, $\varnothing 105 \text{ mm}$, 50 Hz	Kolmeks (2012)
Pump head at q / shut-off head	$H = 14 \text{ m}$ / $H_0 = 14.2 \text{ m}$	Kolmeks (2012), Figure 3
Tank pressure p_{tank}	0.2 barG	Assumption (A2)
Valve candidates	RE-series DN 25 PN40, M1-series DN 25 PN40	Assignment data

Assumptions:

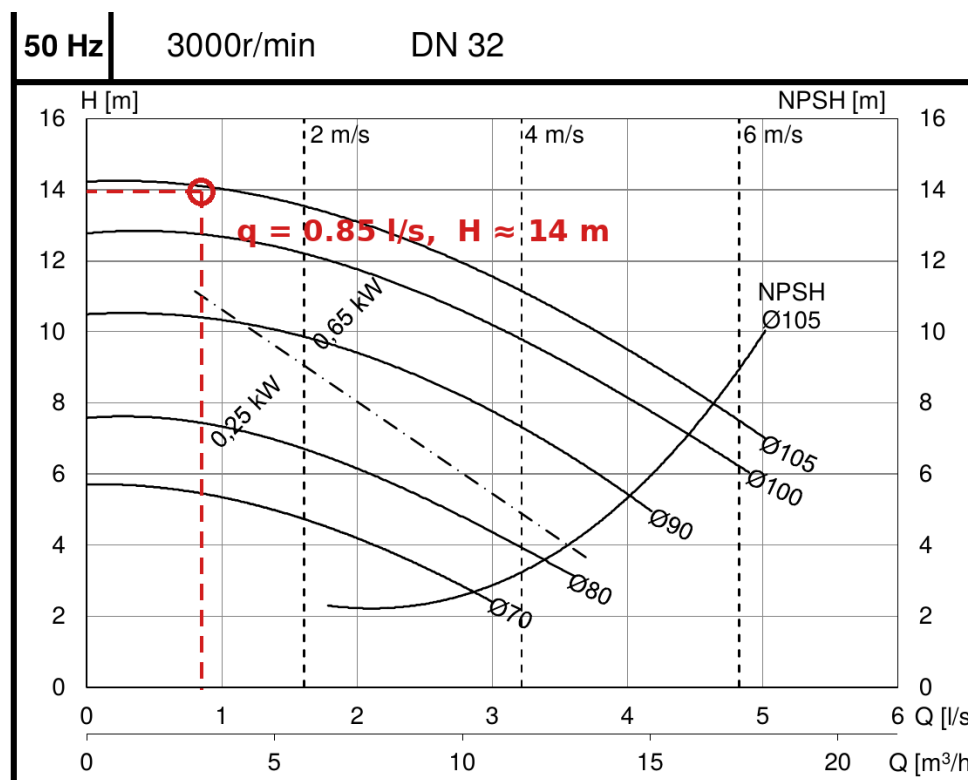
- A1 Only the section pump $\rightarrow$ valve $\rightarrow$ tank is considered, as required by the assignment. Suction-side losses and static head differences are neglected because the tank is both source and sink.
- A2 For the same reason the tank pressure cancels in the pressure balance. $p_{\text{tank}} = 0.2 \text{ barG}$ is not given in the assignment; it only defines the absolute pressure level in Nelprof (margin to the vapour pressure), not Δp_{valve} .
- A3 Water properties at 25°C; $\rho = 1000 \text{ kg/m}^3$ is used (exact value 997 kg/m^3 , difference below 0.3 %).
- A4 The inner diameter equals the nominal size ($d = 32 \text{ mm}$), as in the pressure drop tables of the Flow Control Manual (Valmet, 2022, p. 154) and in Nelprof (pipeline size 32 mm). With a real inner diameter of about 35 mm the velocity would drop to about 0.9 m/s and the friction losses by roughly 30 %; since the total loss is below 2 % of the pump head, the effect on the result is negligible.
- A5 The design flow is the maximum process flow for the DP_m calculation. Δp_0 is the pump shut-off pressure (closed valve, no flow, no losses).

A6 LAA01AA009 (safety valve symbol in the P&I diagram) is a closed side branch in normal operation and causes no flow loss.

3.2 Pump Curve Analysis

According to the manufacturer's performance curve of the Kolmeks L-32A/2 pump with a 105 mm impeller at 50 Hz (Kolmeks, 2012), the pump delivers $H = 14$ m at $q = 0.85$ l/s and $H_0 \approx 14.2$ m at zero flow (Figure 3). The curve is very flat in this range, so the available pressure hardly depends on the flow.

Figure 3. Pump curve of the Kolmeks L-32A/2 (50 Hz) with the operating point $q = 0.85$ l/s, $H \approx 14$ m marked in red



Note. Adapted from Kolmeks (2012).

Converting head to pressure ($\rho = 1000$ kg/m³, $g = 9.81$ m/s²):

$$p_{\text{pump}} = \rho g H = 1000 \cdot 9.81 \cdot 14 = 137\,340 \text{ Pa} \approx 1.37 \text{ bar} \quad (1)$$

$$\Delta p_0 = \rho g H_0 = 1000 \cdot 9.81 \cdot 14.2 = 139\,302 \text{ Pa} \approx 1.39 \text{ bar} \quad (2)$$

The maximum pressure the pump can provide is therefore the shut-off pressure of about 1.39 bar.

3.3 Pressure Loss Between Pump and Valve

All components between the pump discharge and the valve inlet that cause pressure loss were identified from the P&I diagram and the piping on site (Table 2): 2.38 m of straight pipe (segments of 20, 75, 25, 37, 31 and 50 cm), six 90° elbows ($r = 8$ cm), the flowmeter CF202, the pressure transmitter CP202 and the temperature sensor CT202. The manual valves LAA01AA006, LAA01AA101 and LAA01AA007 are located on the suction side of the pump and therefore do not belong to this section.

Flow velocity and Reynolds number in the DN32 pipe:

$$v = \frac{4q}{\pi d^2} = \frac{4 \cdot 0.85 \cdot 10^{-3}}{\pi \cdot 0.032^2} \approx 1.06 \text{ m/s}, \quad Re = \frac{vd}{\nu} = \frac{1.06 \cdot 0.032}{0.893 \cdot 10^{-6}} \approx 3.8 \cdot 10^4 \quad (3)$$

The flow is turbulent. The dynamic pressure is $\rho v^2/2 = 559$ Pa. The friction factor follows from the Haaland equation (Haaland, 1983) with $\varepsilon/d = 0.046/32 = 1.44 \cdot 10^{-3}$:

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left[\left(\frac{\varepsilon/d}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right] \Rightarrow f \approx 0.026 \quad (4)$$

The pressure loss of the straight pipe and of the elbows is

$$\Delta p_{pipe} = f \frac{L}{d} \frac{\rho v^2}{2} = 0.026 \cdot \frac{2.38}{0.032} \cdot 559 \approx 1.08 \text{ kPa} \quad (5)$$

$$\Delta p_{elbows} = n K \frac{\rho v^2}{2} = 6 \cdot 0.25 \cdot 559 \approx 0.84 \text{ kPa} \quad (6)$$

The temperature sensor is the only intrusive element. It is treated as a cylinder in cross flow (drag coefficient $C_D \approx 1.2$) with the well diameter $d_w = 11$ mm from the SKS catalogue (SKS Automaatio, 2015) and a conservative immersion length $L_w = d = 32$ mm (well reaching the opposite wall):

$$\Delta p_{CT202} = C_D \frac{\rho v^2}{2} \frac{d_w L_w}{A} = 1.2 \cdot 559 \cdot \frac{0.011 \cdot 0.032}{8.04 \cdot 10^{-4}} \approx 0.29 \text{ kPa} \quad (7)$$

Table 2. Pressure loss between pump discharge and valve inlet at $q = 0.85$ l/s

No.	Component	Basis	Δp [bar]	Source
1	Straight pipe, 2.38 m, DN 32	$f = 0.026$, $L/d = 74.4$	0.011	Haaland (1983)
2	6× elbow 90°, $r/d = 2.5$	$K = 0.25$ each, $\sum K = 1.5$	0.008	Crane Co. (2009)
3	Flowmeter CF202 (ProcessMaster 300)	Electromagnetic, full bore without internals; $K \approx 0$ (assumed)	≈ 0	ABB (n.d.)
4	Pressure transmitter CP202 (SATRON VT)	Pressure tapping, no through-flow, $K \approx 0$	≈ 0	Satron Instruments (2015)
5	Temperature sensor CT202 (SKS Pt100)	Well $\varnothing 11$ mm, $L_w = 32$ mm, $C_D = 1.2$	0.003	SKS Automaatio (2015)
Total pump → valve			0.022	

As a cross-check, the web calculator *Pressure Drop Online* (Pressure Drop Online, n.d.) gives 0.021 bar for the pipe and the six elbows at $q = 0.85$ l/s (DN 32), compared with 0.019 bar from rows 1 and 2 of Table 2. The total loss of about 0.02 bar is small compared with the pump pressure of 1.37 bar; practically the entire pump pressure remains for the control valve.

Downstream of the valve, the short 0.12 m pipe with one elbow discharges into the tank:

$$\Delta p_{down} = 0.026 \cdot \frac{0.12}{0.032} \cdot 559 + 0.25 \cdot 559 \approx 0.002 \text{ bar} \quad (8)$$

3.4 Pressure Drop Across the Valve and DP_m

Since the static head difference is zero, the pump pressure is dissipated in the pipework and in the valve:

$$\Delta p_{valve} = p_{pump} - \Delta p_{pump \rightarrow valve} - \Delta p_{down} = 1.373 - 0.022 - 0.002 \approx 1.35 \text{ bar} \quad (9)$$

With the assumed tank pressure of 0.2 barG this gives a valve inlet pressure of $0.2 + 1.373 - 0.022 \approx 1.55$ barG and a valve outlet pressure of about 0.2 barG. The inlet pressure is below the design pressure of the existing valve (500 kPa) and far below the PN40 rating of the candidates.

According to the Flow Control Manual (Valmet, 2022, p. 23, eq. 10) the process pressure ratio DP_m is the valve pressure drop at the maximum design flow divided by the pressure drop across the closed valve:

$$DP_m = \frac{\Delta p_m}{\Delta p_0} = \frac{1.35}{1.39} \approx 0.97 \quad (10)$$

This is far above the rule-of-thumb value $DP_m = 0.30$ for liquid piping (Valmet, 2022, p. 30, eq. 15), which assumes that the valve takes about one third of the total dynamic pressure loss. Here the piping is very short with low losses and the pump curve is flat, so the valve takes almost the entire pump

pressure. The consequences for the installed characteristic and gain are discussed in Section 5.6.

3.5 Required Flow Coefficient

$$C_v = \frac{q}{N_1 \sqrt{\Delta p_{valve}/G_f}} = \frac{3.06 \text{ m}^3/\text{h}}{0.865 \cdot \sqrt{1.35/1.0}} \approx 3.04 \quad (11)$$

with $N_1 = 0.865$ for q in m^3/h and Δp in bar, and specific gravity $G_f = 1.0$. Nelprof calculates $C_v = 3.05$ (Section 4); the small difference comes from rounding.

4 Sizing Results (Nelprof)

The process data of Table 3 were entered into Valmet Nelprof (Valmet, n.d.) for two DN 25 PN40 valve alternatives. The complete input and result screens are shown in Appendix 1 (Figures 9 and 10); the results are summarised in Table 4.

Table 3. Nelprof input data

Parameter	Value
Liquid / flow rate	Water / 0.85 l/s
Inlet temperature	25°C
Inlet pressure / outlet pressure	1.55 barG / 0.20 barG
Pressure difference Δp_{valve}	1.35 bar
Vapour pressure	0.0317 bar(a)
Pipe size inlet / outlet, schedule	32 mm / 32 mm, Std

Table 4. Nelprof sizing results at the design operating point

Result	RE-series	M1-series
Valve, size, pressure class	Segment valve, DN 25, PN40	Ball valve, DN 25, PN40
Trim / maximum capacity	Reduction 2 / 5 C_v	Full bore / 105 C_v
Required capacity C_v	3.05	3.05
Travel / opening angle	82.8 % / 80.5°	31.1 % / 39.0°
Recommended travel limits	3–95 %	5–90 %
Noise at operating point	50 dBA	41 dBA
Valve outlet / pipe outlet velocity	1.73 m/s / 1.06 m/s	1.73 m/s / 1.06 m/s
Choked Δp	2.11 bar	2.03 bar
Pressure recovery factor F_L	0.91	0.90
Sizing result status	ok	ok

Note. Values from the Nelprof result screens (Valmet, n.d.), Figures 9 and 10.

All values in Table 4 refer to the design operating point and therefore depend only on the entered flow and pressures, not on the process model used for the installed charts.

4.1 Alternative 1: RE-Series Segmented Ball Valve

According to the manufacturer's documentation, the RE-series is a V-port segment valve with an equal-percentage inherent characteristic. For DN 25 a reduced C_v trim is available to control very low flows (Valmet, 2025). The reduced trim (reduction 2, maximum capacity $5 C_v$) was selected to match the low required capacity. The required $C_v = 3.05$ agrees with the hand calculation (3.04). The valve operates at 82.8 % travel, which is inside the recommended travel limits of 3–95 % and uses most of the available stroke. The choked pressure drop of 2.11 bar is clearly above the actual $\Delta p = 1.35$ bar, so the flow is not choked and no heavy cavitation is expected.

4.2 Alternative 2: M1-Series Standard Ball Valve

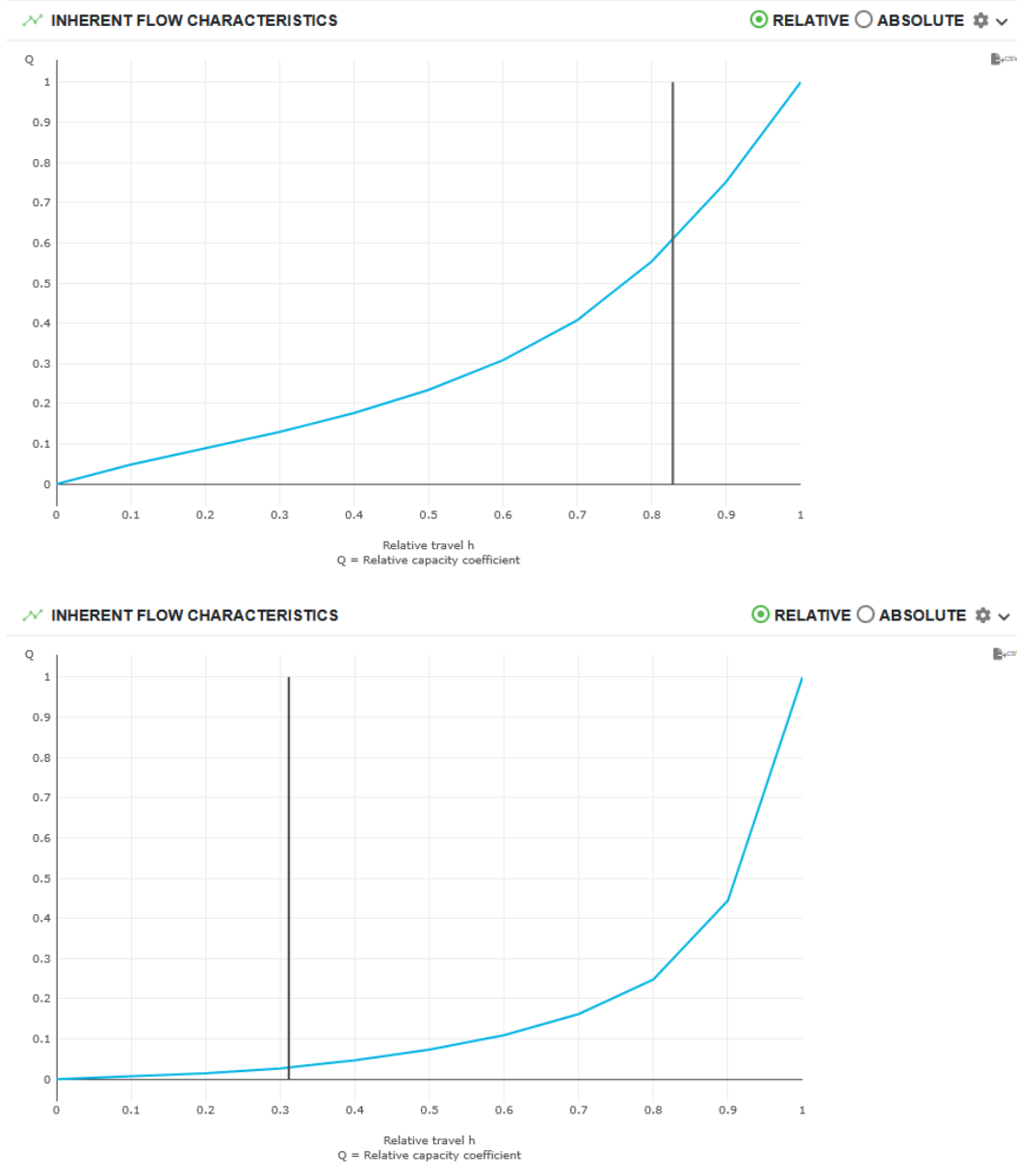
The standard M1-series ball valve (maximum capacity $105 C_v$) was sized for the same process conditions. It reaches the required $C_v = 3.05$ at only 31.1 % travel (39.0°), i.e. it uses about 3 % of its capacity. The noise at the operating point is low (41 dBA) and the choked pressure drop (2.03 bar) is also above 1.35 bar. The valve outlet velocity of 1.73 m/s is the same for both valves (same DN 25 outlet) and far below the recommended maximum inlet velocity of 10 m/s for segment and ball valves in continuous duty (Valmet, 2022, p. 65).

5 Graphical Analysis (Nelprof Charts)

Figures 4 to 8 show the Nelprof charts; in each figure the RE-series is displayed on top and the M1-series below. The vertical line marks the design operating point. The installed charts (Figures 5 to 8) were calculated by Nelprof with its default process model, which corresponds to $DP_m \approx 0.30$ and not to the calculated $DP_m = 0.97$ (proof in Section 5.4). Sections 5.1 to 5.5 describe the charts as calculated; the effect of the real DP_m is evaluated in Section 5.6.

5.1 Inherent Flow Characteristics

Figure 4. Inherent flow characteristics (top: RE-series, bottom: M1-series)

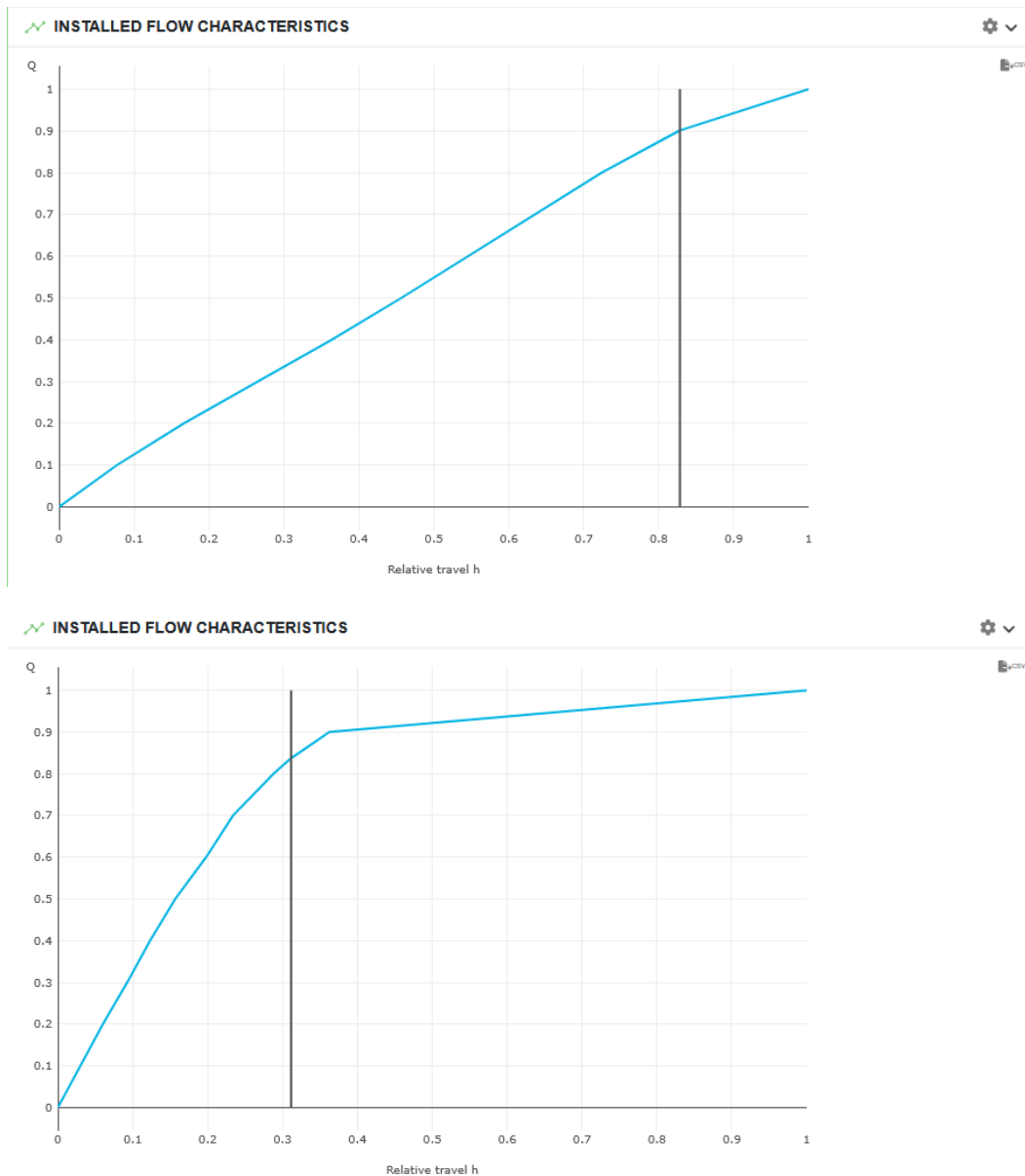


Note. Screenshots from Nelprof (Valmet, n.d.).

Figure 4 shows the relative capacity of the valve as a function of relative travel at constant pressure drop. The RE-series (top) has the equal-percentage characteristic stated by the manufacturer (Valmet, 2025): at 50 % travel the relative capacity is about 23 %, at 80 % travel about 55 %. The M1-series (bottom) has a total capacity of $105 C_v$ and is strongly oversized for a required capacity of about $3 C_v$. Its curve is flat over the first two thirds of the travel and rises steeply only close to the fully open position. The required capacity is therefore reached at a small opening of about one third of the travel, and most of the capacity of the valve is never used.

5.2 Installed Flow Characteristics

Figure 5. Installed flow characteristics (top: RE-series, bottom: M1-series)



Note. Screenshots from Nelprof (Valmet, n.d.); calculated with the Nelprof default process model ($DP_m \approx 0.30$).

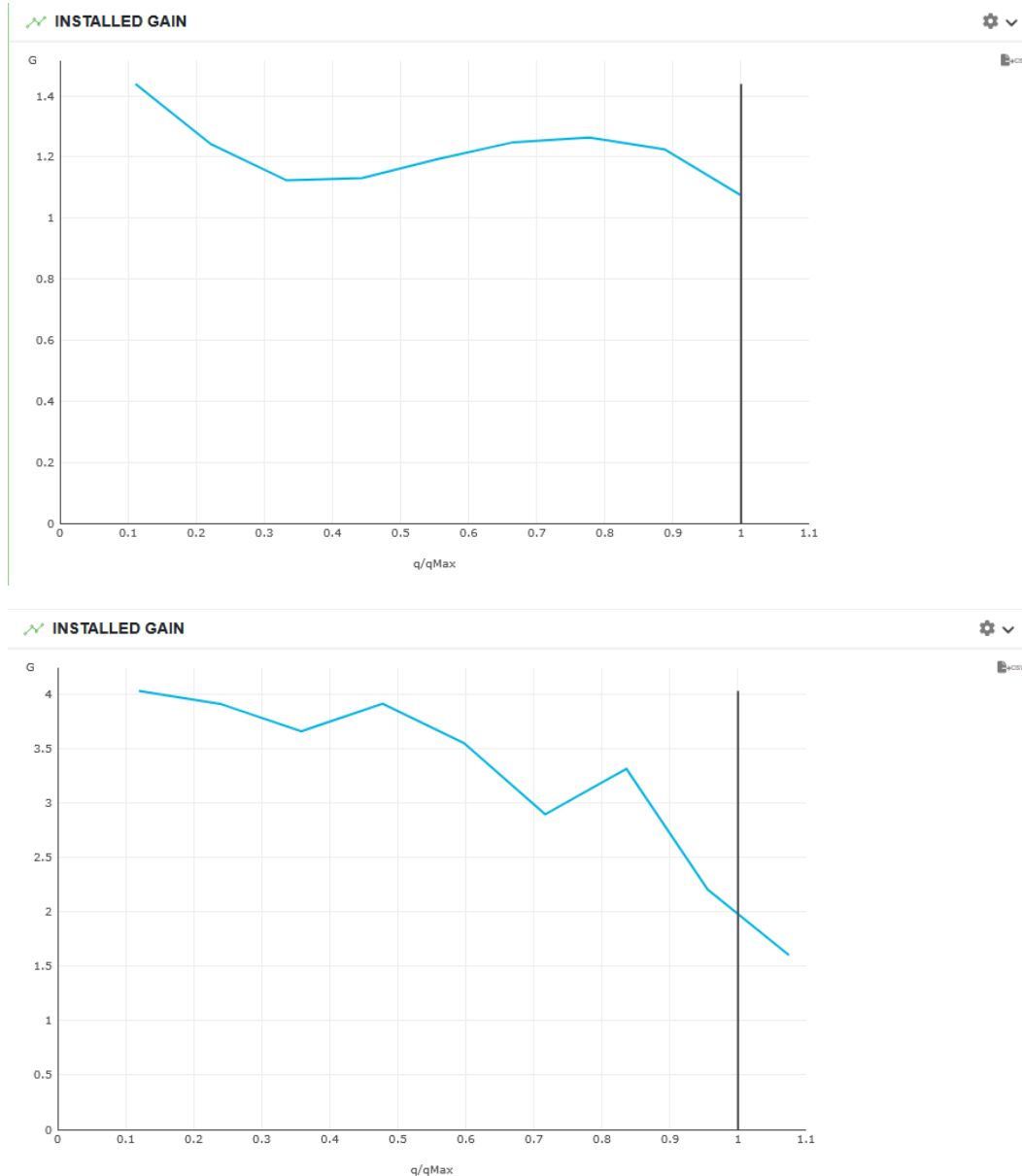
The installed characteristic in Figure 5 includes the pressure behaviour of the process. The relative flow $Q = q/q_f$ is the flow divided by the flow through the fully open valve (Valmet, 2022, p. 28). For the RE-series (top) the curve is almost linear: travel 0.5 gives a relative flow of about 0.55 and travel 0.8 about 0.87. At the operating point (82.8 % travel) the relative flow is about 0.90, i.e. the fully open valve would pass about $0.85/0.90 \approx 0.94$ l/s. In this model the RE valve therefore has a flow reserve of only about 10 %.

The oversized M1-series (bottom) reaches 80 % of its installed flow at about 28 % travel and 90 % at about 36 % travel. Above that the curve is almost flat, because the piping pressure loss in the model

then limits the flow. The usable control range is compressed into the first third of the travel and the remaining travel is of no use for this process.

5.3 Installed Gain

Figure 6. Installed gain (top: RE-series, bottom: M1-series)



Note. Screenshots from Nelprof (Valmet, n.d.); calculated with the Nelprof default process model ($DP_m \approx 0.30$).

The installed gain $G = dQ_p/dh$ describes the sensitivity of the flow to a change in valve position. In Nelprof it is plotted over q/q_m , the flow relative to the maximum process flow (Valmet, 2022, p. 28). According to the Flow Control Manual (Valmet, 2022, p. 25, eq. 13 and 14), the change in gain in the operating range should satisfy $G_{max}/G_{min} < 2$ and the gain should not fall below 0.5.

For the RE-series (Figure 6, top) the gain varies between about 1.07 at the operating point and 1.44

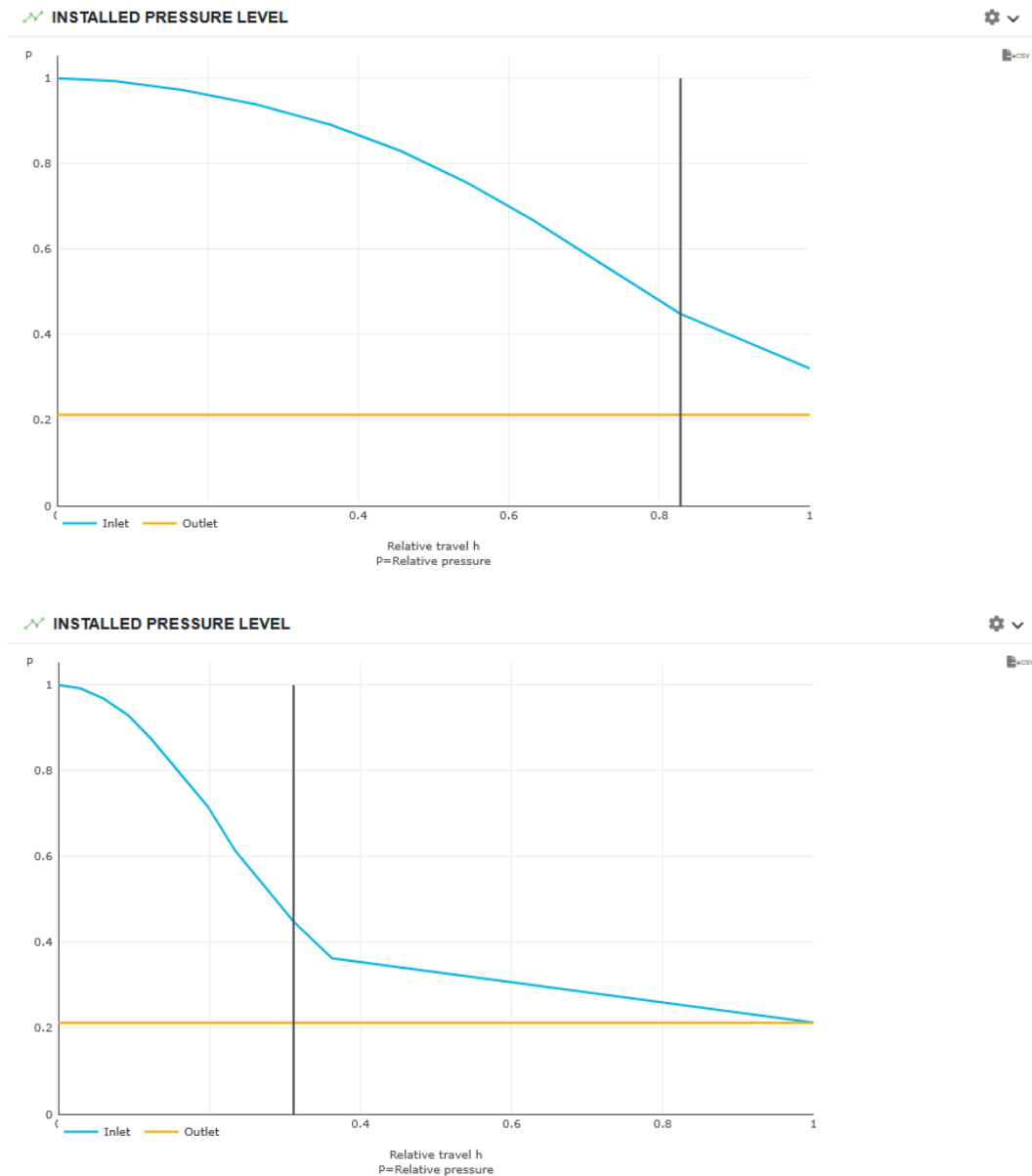
at low flow, so $G_{max}/G_{min} \approx 1.35$. Both criteria are met and the gain is close to 1, i.e. a 1 % change in travel changes the flow by about 1 % of the maximum flow.

For the M1-series (bottom) the gain lies between about 2.0 at the operating point and 4.0 at low flow, so $G_{max}/G_{min} \approx 2.0$, which is exactly at the limit of eq. 13. In addition, the gain is at a high absolute level: a 1 % change in travel changes the flow by 2–4 %. Since the relative flow error of a control valve is the gain multiplied by the travel error (Valmet, 2022, p. 25), every positioning error of the valve is amplified by a factor of up to four. This promotes oscillation of the control loop and makes tuning of the controller more difficult.

The two curves match the examples in the Flow Control Manual: the M1 gain corresponds to the oversized valve in Figure 19 of the manual (high gain at small flows, poor accuracy and possible hunting), while the RE gain corresponds to the properly sized valve in Figure 20 of the manual with a lower and more constant gain (Valmet, 2022, p. 26).

5.4 Installed Pressure Level

Figure 7. Installed pressure level (top: RE-series, bottom: M1-series)



Note. Screenshots from Nelprof (Valmet, n.d.); calculated with the Nelprof default process model ($DP_m \approx 0.30$).

Figure 7 shows the absolute valve inlet and outlet pressure relative to the inlet pressure of the closed valve. With increasing opening the inlet pressure falls from 1.0 to about 0.45 at the operating point for both valves and further to about 0.32 (RE) and 0.21 (M1) at the fully open position. The outlet pressure stays constant at about 0.21. For the M1 the inlet and outlet curves meet at full opening, which means that the valve no longer throttles and the whole pressure is lost in the piping of the model.

The chart also shows which process model Nelprof used. At the operating point the inlet pressure is $1.55 \text{ barG} = 2.56 \text{ bar(a)}$ and its relative value is 0.45, so the closed-valve inlet pressure in the model

is $2.56/0.45 \approx 5.7 \text{ bar(a)}$. This is confirmed by the outlet: $1.21 \text{ bar(a)}/5.7 \text{ bar(a)} \approx 0.21$. Thus

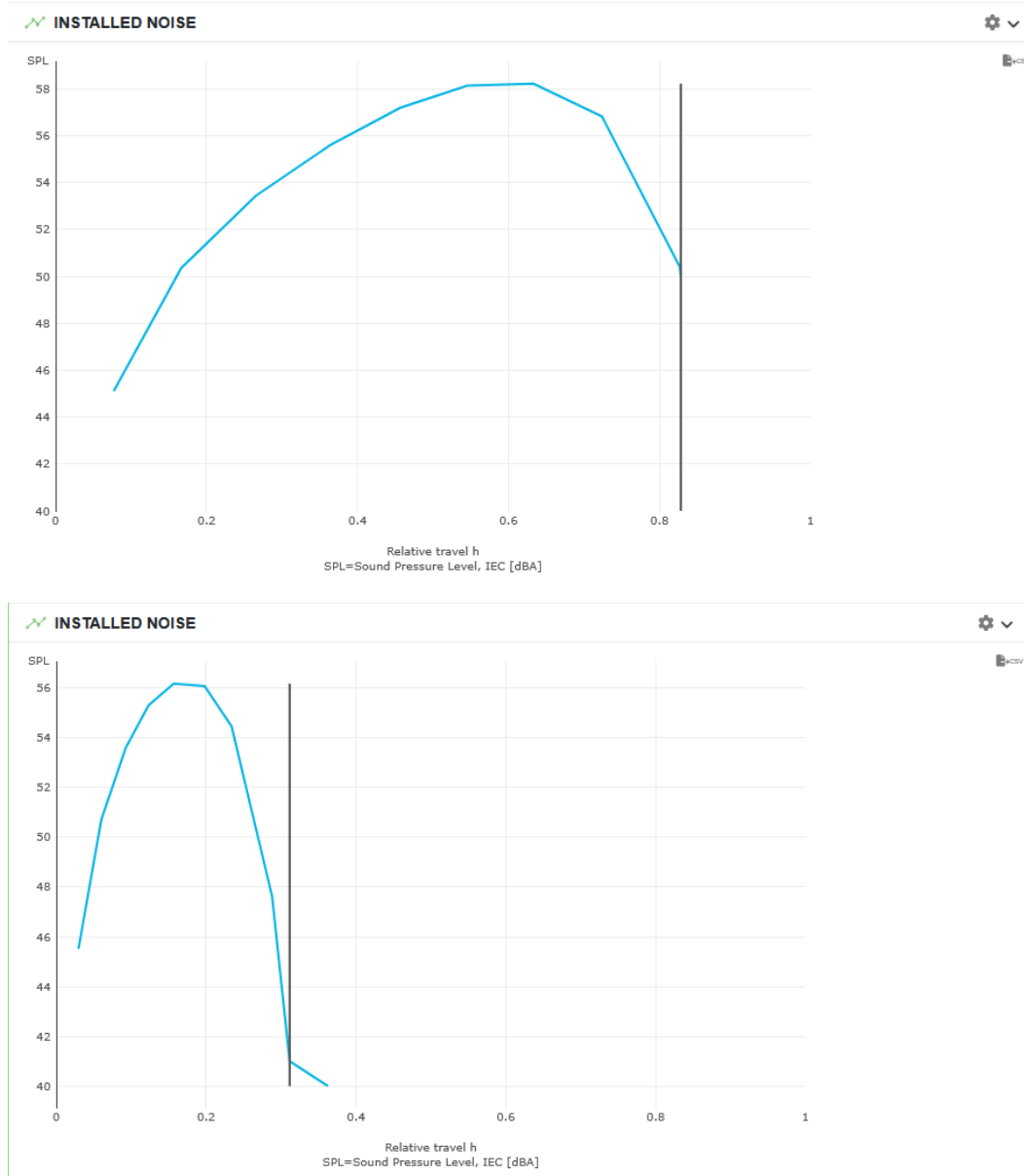
$$\Delta p_{0,\text{Nelprof}} \approx 5.7 - 1.21 \approx 4.5 \text{ bar}, \quad DP_{m,\text{Nelprof}} \approx \frac{1.35}{4.5} \approx 0.30 \quad (12)$$

Nelprof therefore used the default value $DP_m \approx 0.30$ (Valmet, 2022, p. 30) instead of the calculated 0.97, because no second operating point and no DP_m value had been entered. In the real process the inlet pressure can only rise to about 1.59 barG (pump shut-off pressure plus tank pressure) when the valve closes.

In both cases the pressure downstream of the valve stays far above the vapour pressure of 0.0317 bar(a), so flashing does not occur (Valmet, 2022, p. 53).

5.5 Installed Noise

Figure 8. Installed hydrodynamic noise (top: RE-series, bottom: M1-series)



Note. Screenshots from Nelprof (Valmet, n.d.); calculated with the Nelprof default process model ($DP_m \approx 0.30$).

The predicted hydrodynamic noise is a good indicator of cavitation intensity (Valmet, 2022, p. 52). In Figure 8 the RE-series peaks at about 58 dBA at 55–65 % travel and reaches 50 dBA at the operating point. The M1-series peaks at about 56 dBA at 15–20 % travel and reaches 41 dBA at the operating point. The noise limit for preventing cavitation damage in valves up to DN 80 is 80 dB(A) (Valmet, 2022, p. 53, Table 1). Both valves stay more than 20 dB below this limit, and $\Delta p_{valve} = 1.35$ bar is below the choked pressure drop of both valves. Neither cavitation damage nor flashing is expected. The noise at partial load would be somewhat different with the real DP_m , but because the whole pressure difference is only about 1.4 bar, the margin to the limit remains large.

5.6 Influence of the Real Process Pressure Ratio

The installed charts above were calculated with $DP_m \approx 0.30$, whereas the calculated value for this process is $DP_m = 0.97$. With DP_m close to 1 the pressure drop across the valve stays almost constant over the whole flow range, so the installed characteristic becomes almost identical to the inherent characteristic (Valmet, 2022, pp. 22–24). The installed curves in Figure 5 are therefore more linear than they would be in reality.

For the RE-series this means that the installed characteristic would be nearly equal-percentage, like the top curve in Figure 4. The gain can be estimated from the slope of the inherent curve, divided by the relative capacity at the operating point (about 0.61): between 10 % and 30 % travel the slope is about 0.4, which gives $G \approx 0.4/0.61 \approx 0.65$; near the operating point the slope is about 2.0, which gives $G \approx 3.3$. The gain would thus change by a factor of about five and the criterion $G_{max}/G_{min} < 2$ (Valmet, 2022, p. 25) would not be met. For the M1-series the same estimate gives a gain of roughly 5–7 at the operating point (slope about 0.2 at a relative capacity of only 0.03), i.e. even higher than in Figure 6.

On the other hand, the flow reserve of the RE valve is larger than the 10 % read from Figure 5: since the pressure drop hardly decreases when the valve opens, the fully open valve ($5 C_v$) could pass roughly $5/3.05 \approx 1.6$ times the required capacity, reduced somewhat by the pump curve and the piping losses at higher flow. The 10 % value is therefore a conservative estimate.

The non-linear gain of the RE valve can be compensated by characterization, either in the ND9000 digital valve controller or in the controller output, as described in the Flow Control Manual (Valmet, 2022, pp. 32–34). With such a correction the installed characteristic becomes approximately linear. For the M1 valve characterization does not help much, because the whole control range is compressed into a small part of the stroke, where travel errors are amplified strongly.

6 Comparative Evaluation and Conclusions

6.1 Valve Performance Comparison

Table 5 summarises the comparison of the two alternatives.

The M1-series ball valve, which corresponds to the currently installed valve type, offers tight shut-off and a very high maximum capacity. For a throttling duty at about $3 C_v$ this capacity is a disadvantage (JAMK University of Applied Sciences, 2024): the valve uses only about one third of its stroke, reaches 90 % of its installed flow at about 36 % travel and has a high installed gain of up to 4. This leaves little resolution for the controller.

The RE-series segmented ball valve with reduced trim matches the required capacity much better. It uses most of its stroke and has an installed gain close to 1 that fulfils the criteria of the Flow Control

Table 5. Comparison of the RE-series and M1-series valves

Criterion	RE-series (reduced trim, $5 C_v$)	M1-series ($105 C_v$)
Travel / opening at design flow	82.8 % / 80.5°	31.1 % / 39.0°
Use of capacity	about 60 %	about 3 %
Installed gain (Nelprof)	1.07–1.44, $G_{max}/G_{min} \approx 1.35$ (criterion met)	2.0–4.0, $G_{max}/G_{min} \approx 2.0$ (at the limit)
Noise at operating point / peak	50 dBA / 58 dBA	41 dBA / 56 dBA
Choked Δp vs. $\Delta p = 1.35$ bar	2.11 bar, not choked	2.03 bar, not choked
Cavitation / flashing	no / no	no / no
Flow reserve	about 10 % (Nelprof model), larger with real DP_m	large, but in the flat part of the curve

Manual. Two limitations have to be noted. First, in the Nelprof model the flow reserve at the design flow is only about 10 %; with the real DP_m the reserve is larger, but a flow requirement clearly above 0.85 l/s should be checked again. Second, because of the real $DP_m \approx 0.97$ the installed characteristic will be close to equal-percentage and the gain will vary more than in Figure 6; a characterization in the valve controller or in the control system is therefore recommended (Section 5.6).

6.2 Product Documentation of the Selected Valve

The manufacturer's documentation (Valmet, 2025) describes the RE-series as an integrally flanged V-port segment valve in quarter-turn design with an equal-percentage inherent characteristic. The reduced C_v trim is available for DN 25 valves only and is intended for very low flows, which matches the required capacity of about $3 C_v$. Standard units are supplied with pneumatic diaphragm or cylinder actuators and ND9000 intelligent valve controllers, which also offer the characterization function mentioned in Section 5.6.

6.3 Actuator

The installed AUMA SGR 04.3 is an electric part-turn actuator. The manufacturer's standard configuration for the RE-series uses a pneumatic actuator with an ND9000 valve controller (Valmet, 2025). Changing the actuator is a secondary recommendation; the valve selection does not depend on it.

6.4 Final Verdict

Based on the process data, the pressure loss calculation between pump and valve ($\Delta p_{valve} = 1.35$ bar, $DP_m = 0.97$, $C_v \approx 3.05$) and the Nelprof analysis, the **RE-series segmented ball valve (DN 25, PN40, reduced trim with $5 C_v$)** is more suitable for this application than the M1-series ball valve. It has a suitable capacity, uses most of its stroke and has a stable installed gain, without

cavitation or flashing. The M1-series is strongly oversized, uses only about one third of its stroke and has a high installed gain. For the RE valve a characterization in the ND9000 or in the controller is recommended, and the limited flow reserve should be kept in mind if the flow requirement increases.

Declaration on the Use of Artificial Intelligence

In accordance with the JAMK guidelines, generative AI (Claude, Anthropic) was used as a support tool. It was used for language checking, for formatting the LaTeX document according to the JAMK reporting template and for reviewing the calculation approach (pressure balance, DP_m definition) and the consistency between the calculations, the Nelprof results and the text. The pipe friction calculation, the component losses, the pressure drop across the valve, DP_m and C_v (Section 3) were calculated and checked by the author; the pipe loss was cross-checked with *Pressure Drop Online*. All sizing was performed by the author in Nelprof, the values read from the Nelprof charts were checked by the author, and the interpretation of the results and the valve selection are the author's own. The author is fully responsible for the technical content.

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Appendices

Appendix 1. Nelprof Setup and Results

Figure 9. Nelprof input data and results, RE-series

Valmet

Nelprof

Mattiias Lampert | Log out

Autosave

Tag

SIZING NAME

PHASE

LIQUID

DESCRIPTION

COMMENTS

Calculate

Save

Type search term

Filter

LAA01AA202

Area

Tag

Current Valve

New Valve

SIZING DATA

3rd party valve

Resistor

Actuator

FLUID PROPERTIES

UNIT

VALUE

Liquid Specific Gravity

1

Critical pressure

atmA

221.2

Design pressure min

barG

Design pressure max

barG

Design temperature min

Celsius

Design temperature max

Celsius

PIPELINE

UNIT

VALUE

Schedule

mm

Std

Inlet size

mm

32 mm

Outlet size

mm

32 mm

Insulation type

No insulation

VALVE

VALUE

Type

Segment Valve

Pressure class

EN PN40

Series

RE

Size

Automatic

Size reduction

Automatic

Recommended valve travel limits

3

95

Design style

Not applicable

Flow direction

Seat code

S

Trim characteristics

Trim option

General

PROCESS DATA

1. Case

2. Case

3. Case

4. Case

Liquid flow rate

l/s

0.85

Inlet temperature

Celsius

25

Inlet pressure

barG

1.55

Pressure difference

bar

1.35

Outlet pressure

barG

0.2

Vapour pressure

barA

0.0317

Dynamic viscosity

cp

RESULTS

RE | EN PN40 | SIZE: 25 MM / 1" | REDUCTION: 2 | MAX CAPACITY: 5 CV

25 mm / 1" | 40 mm / 1.5"

RESULT

UNIT

1. CASE

2. CASE

3. CASE

4. CASE

Required Capacity

Cv

3.05

Travel

%

82.8

Opening

°

80.5

Noise

dBA

50

Valve outlet velocity

m/s

1.73

Pipe outlet velocity

m/s

1.06

Choked Dp

bar

2.11

FL

0.91

Sizing result status

ok

CHARTS

SIGMA VALUES

Note. Screenshot from Nelprof (Valmet, n.d.).

Figure 10. Nelprof input data and results, M1-series

Valmet

Nelprof

Mattiias Lampert | Log out

Autosave

Tag

SIZING NAME

PHASE

LIQUID

DESCRIPTION

COMMENTS

Calculate

Save

Type search term

Filter

LAA01AA202

Area

Tag

Current Valve

New Valve

SIZING DATA

3rd party valve

Resistor

Actuator

FLUID PROPERTIES

UNIT

VALUE

Liquid Specific Gravity

1

Critical pressure

atmA

221.2

Design pressure min

barG

Design pressure max

barG

Design temperature min

Celsius

Design temperature max

Celsius

PIPELINE

UNIT

VALUE

Schedule

mm

Std

Inlet size

mm

32 mm

Outlet size

mm

32 mm

Insulation type

No insulation

VALVE

VALUE

Type

Ball Valve

Pressure class

EN PN40

Series

M1

Size

Automatic

Size reduction

Automatic

Recommended valve travel limits

5

90

Design style

Seat supported

Flow direction

Seat code

T

Trim characteristics

Trim option

General

PROCESS DATA

1. Case

2. Case

3. Case

4. Case

Liquid flow rate

l/s

0.85

Inlet temperature

Celsius

25

Inlet pressure

barG

1.55

Pressure difference

bar

1.35

Outlet pressure

barG

0.2

Vapour pressure

barA

0.0317

Dynamic viscosity

cp

RESULTS

M1 | EN PN40 | SIZE: 25 MM / 1" | REDUCTION: 0 | MAX CAPACITY: 105 CV

25 mm / 1" | 40 mm / 1.5"

RESULT

UNIT

1. CASE

2. CASE

3. CASE

4. CASE

Required Capacity

Cv

3.05

Travel

%

31.1

Opening

°

39.0

Noise

dBA

41

Valve outlet velocity

m/s

1.73

Pipe outlet velocity

m/s

1.06

Choked Dp

bar

2.03

FL

0.9

Sizing result status

ok

CHARTS

SIGMA VALUES

Note. Screenshot from Nelprof (Valmet, n.d.).